Reg. No. | | | | | | | | | | | [18MFT3001]

THIRD SEMESTER M.Sc. UNIVERSITY EXAMINATION, NOVEMBER - 2019

SCIENCE OF BEVERAGES

Time:03 Hours **Maximum Marks:70**

PART - A

1. Answer ALL the following questions. **10x2=20**

a. What are bound and free water?

b. What are the different contaminants that affect water quality?

c. What is the permissible limit of Iron in drinking water

d. Name any colorants and acidulants used in beverages?

e. What are the different chemical constituents used in softening of water?

f. Mention the different biological activities of cocoa.

g. Explain sedimentation and coagulation.

h. What are the different steps involved in the processing of beverages.

i. List the microbiological hazards associated with alcoholic beverages.

j. What is water activity?

PART - B

2. Answer THREE of the following questions. **3x6=18**

a. With a neat structure list the properties of water.

b. Explain sorption phenomenon and sorption isotherm.

c. Write a note on boiler feed water treatment.

d. Write a note on health risks of consuming contaminated water.

e. Explain the significance of water quality in food samples.

PART - C

3. Answer FOUR of the following questions. **4x8=32**

a. Outline the drinking water standards as per BSI.

b. Paraphrase on the drinking water standards in India.

c. Explain the processing of various types of tea and coffee leaves.

d. Write a note on FSSAI specifications for beverages.

e. What are preservatives? Explain the role of preservatives and antioxidants in soft drinks.

f. What are the methods adopted to remove iron and manganese from water?

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